

Group meeting

UFSD
literature
reading

From silicon sensor signals to UFSD timing

UFSD literature reading

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Internship update for Prof. Prabhakar Palni's
group

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Goal for today:
showing the theory I learned

Outline

Theory from the book

1. Silicon sensor basics
2. Energy deposition and Landau fluctuations
3. Signal formation and Shockley–Ramo
4. Radiation damage
5. LGAD gain layer and UFSD signal
6. Timing terms and readout ideas

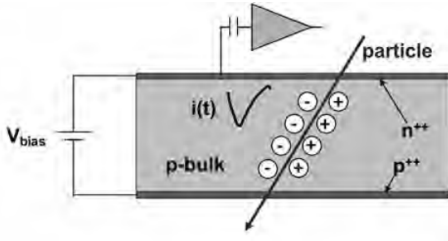
Papers

1. Cartiglia et al., JINST 2014
Timing performance of UFSDs
2. Yang et al., 2024
4H-SiC LGAD characterization

Next

- ▶ Finish the remaining reading.
- ▶ Learn Victory Device more carefully.

Chapter 1: one silicon sensor picture



1 Schematic cross section of an *n-in-p* silicon sensor pad, traversed by an

Basic chain

1. Reverse bias makes a depleted active region.
2. A particle creates electron–hole pairs.
3. The field moves the carriers.
4. Their motion induces the current signal.

$$V_{FD} \propto d^2$$

Why start here UFSD timing still starts from the same sensor signal picture.

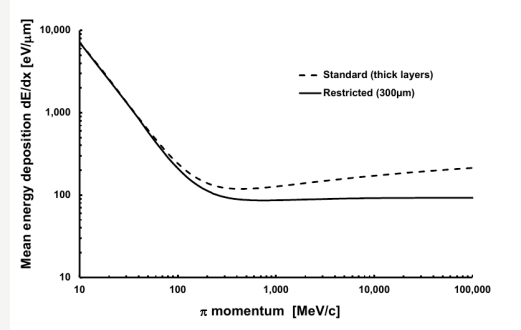
Chapter 1: a detector starts with energy deposition

What I took from the chapter

- ▶ A minimum-ionizing particle gives the reference signal scale.
- ▶ Thin silicon layers collect less charge, and the deposited energy is not a fixed number.
- ▶ The Landau tail matters for timing because event-to-event signal shapes and amplitudes vary.

$$N_{e-h} \approx \frac{\Delta E}{3.6 \text{ eV}}$$

thin sensor $\Rightarrow$ smaller signal and larger fluctuations

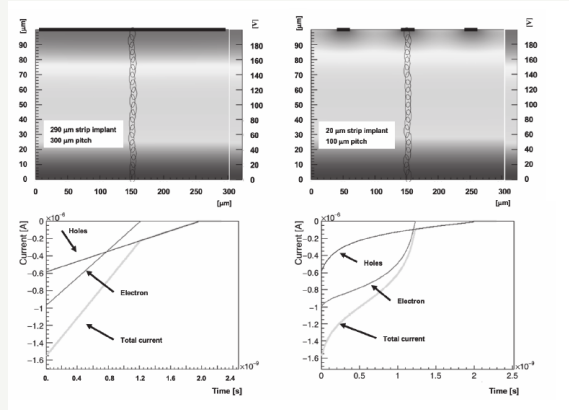


Chapter 1: signal formation is not just charge collection

Shockley–Ramo

$$i_k(t) = -q \vec{v}(t) \cdot \vec{E}_w$$

- ▶ The signal is induced while carriers move.
- ▶ The weighting field is a geometry problem, not the same as the bias field.
- ▶ Large pad-like geometries keep the response more uniform than narrow structures.



Chapter 1: radiation damage changes the sensor problem

Surface damage

trapped interface charge and segmentation concerns



Bulk damage

leakage current, trapping, and effective doping changes



Why LGADs care

acceptor removal weakens the gain implant

Two channels

Simple takeaway

- ▶ More leakage current raises the noise problem.
- ▶ Trapping reduces the useful collected charge.
- ▶ If the gain implant loses active acceptors, the LGAD gain drops.

Radiation hardness For a gain device, it matters whether the high-field gain region still works after irradiation.

Chapter 2: LGAD adds a controlled high-field region

PIN to LGAD

- ▶ UFSDs use the LGAD idea: low gain inside a thin sensor.
- ▶ The gain implant sits near the junction and raises the local field.
- ▶ The design target is useful multiplication without early breakdown.

Device	Gain	Regime
PIN	1	linear
LGAD	10–30	linear

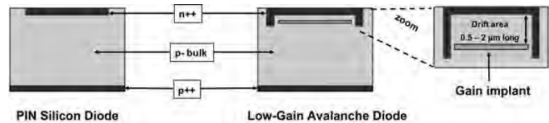


Figure 2.1 A schematic view of: (left) an *n-in-p* PIN diode; (right) a Low-Gain Avalanche Diode. The LGAD design is characterized by the presence of an additional p^+ implant underneath the *pn* junction.

Design knob Gain depends on the electric field and the doping profile.

Chapter 2: the gain implant shape matters

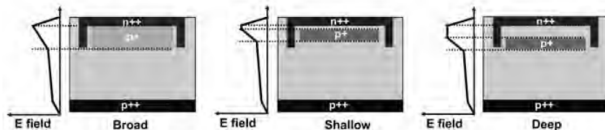


Figure 2.3 Schematic cross cut of a UFSD with broad (*left*), shallow (*middle*), and deep (*right*) gain implants and their respective electric field profiles.

What I learned

- ▶ Broad, shallow, and deep gain implants give different field shapes.
- ▶ The same gain can be more or less sensitive to small design changes.
- ▶ This is one reason doping profiles matter in LGAD work.

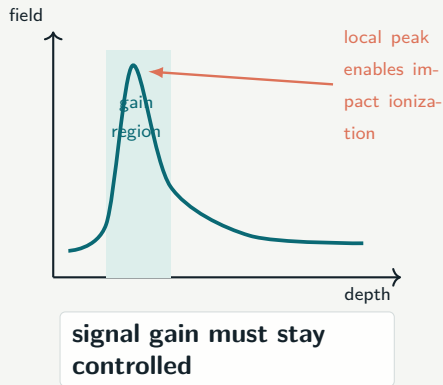
UFSD design The book describes UFSDs as LGADs tuned for timing: thin signal path, uniform weighting field, high drift speed, and high fill factor.

Chapter 2: why the gain layer helps timing

Controlled multiplication

$$N(d) = N(0)e^{\alpha d}$$

- ▶ High field lets carriers create secondary electron-hole pairs by impact ionization.
- ▶ In the low-gain operating window, multiplication increases signal before noise wins.
- ▶ Too little gain gives weak timing leverage; too much field pushes toward breakdown.



Chapter 2: there is a useful gain window

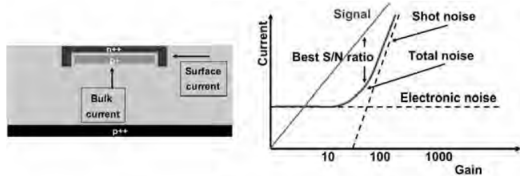


Figure 2.8 *Left:* schematic representation of a UFSD, with the localization of the bulk and of the surface leakage currents. *Right:* signal and shot noise growth as a function of the sensor internal gain.

What the figure says

- ▶ Signal rises with gain.
- ▶ Shot noise also rises with gain.
- ▶ So there is a practical region where the signal-to-noise ratio is best.

Why it matters For LGAD work, gain is a design knob, not a number to maximize blindly.

Chapter 2: UFSD signal formation gives the timing lever

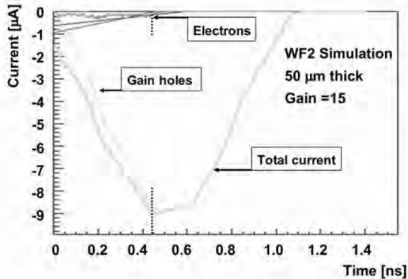


Figure 2.5 WF2 simulation of the total current, with the various contributions due to primary electrons/holes and gain electrons/holes, generated by a MIP traversing a 50 μm-thick UF. The current increases for the duration of the electron current.

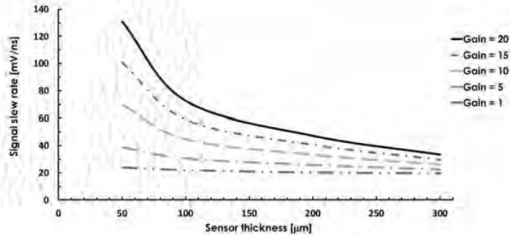
Main relation

$$\frac{dI}{dt} \propto \frac{dV}{dt} \propto \frac{G}{d}$$

- ▶ G : internal gain steepens the signal.
- ▶ d : a thinner active layer shortens the relevant drift scale.
- ▶ Timing improves when the rising edge is steep enough for the front end to locate it precisely.

For timing A steeper rising edge makes the hit time easier to locate.

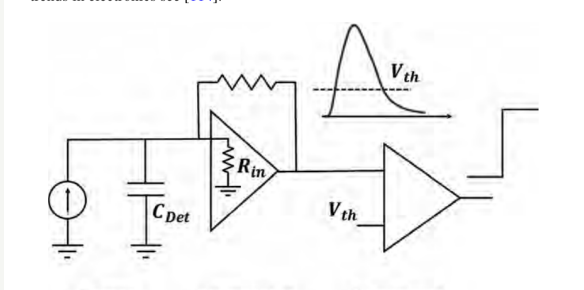
Chapter 2: slew rate rewards thin sensors



One direct trend

- ▶ For a fixed gain, thicker sensors give a smaller slew rate.
- ▶ For a fixed thickness, more gain gives a steeper edge.
- ▶ This is one clean reason thin UFSDs help timing.

Chapter 2: a timing measurement is a chain



Blocks

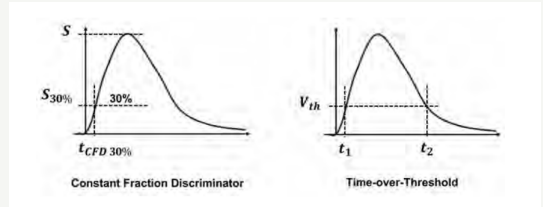
- ▶ The sensor gives a current pulse.
- ▶ The preamplifier shapes it.
- ▶ A comparator defines a crossing time.
- ▶ The TDC digitizes that timing point.

Chapter 2: time walk needs a correction

Why it appears

- ▶ MIP signal amplitudes are not all equal.
- ▶ With a fixed threshold, a larger pulse crosses earlier.
- ▶ That shifts the assigned hit time even when the particle arrival was the same.

CFD	use a fixed fraction of the pulse
ToT	correct using threshold crossing times



Chapter 2: the CFD fraction is also a choice

Ultra-Fast Silicon Detectors

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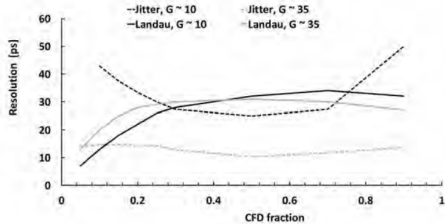


Figure 2.18 WF2 simulation of the jitter and Landau noise terms as a function of the CFD

Reading point

- ▶ The jitter and Landau terms do not move in the same way as CFD fraction changes.
- ▶ So the timing pick-off is a tuning problem, not just a fixed recipe.
- ▶ The best point depends on gain and pulse shape.

Why useful Even if the detector is good, the chosen timing method still changes the final resolution.

Chapter 2: the timing budget has different kinds of error

Time-tagging vocabulary

$$\sigma_t^2 = \sigma_{\text{jitter}}^2 + \sigma_{\text{Landau}}^2 + \sigma_{\text{time walk}}^2 + \sigma_{\text{TDC}}^2$$

Jitter

Noise moves the threshold-crossing time.

Landau

Ionization fluctuations change the signal itself.

Time walk

Larger pulses cross a fixed threshold earlier.

TDC

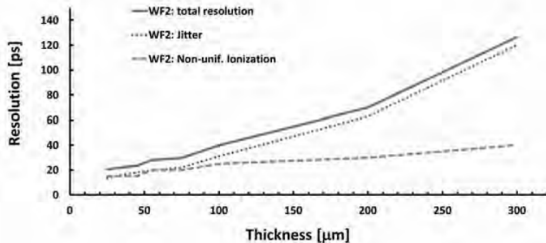
Digitization adds a finite binning term.

$$\sigma_{\text{jitter}} = \frac{N}{dV/dt} \approx \frac{t_r}{S/N}$$

$$\sigma_{\text{time walk}} = \left[\frac{t_r V_{\text{th}}}{S} \right]_{\text{RMS}}$$

$$\sigma_{\text{TDC}} = \frac{\text{TDC}_{\text{bin}}}{\sqrt{12}}$$

Chapter 2: temporal resolution points to thin UFSDs



What the plot says

- ▶ The total timing resolution worsens as thickness grows.
- ▶ Jitter and non-uniform ionization both matter around the thin-sensor region.
- ▶ A 50 μm UFSD is already in the 30–40 ps range in this example.

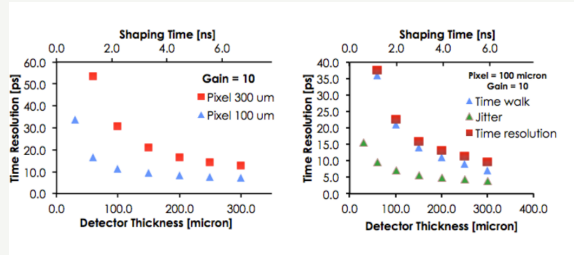
Book point For thin UFSDs, the Landau term becomes one of the main limits.

Paper 1: Cartiglia et al. gives the timing argument

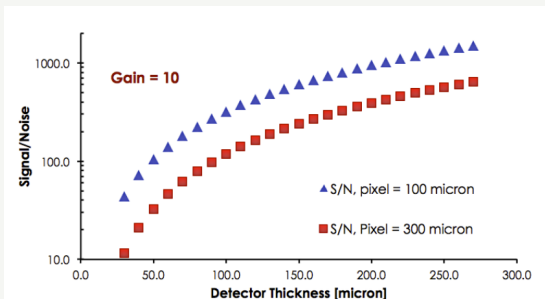
Performance of UFSDs

- ▶ Starts from the time-tagging detector view: threshold crossing, time walk, jitter, TDC.
- ▶ Parameterizes how detector thickness, pixel size, capacitance, and shaping time enter timing.
- ▶ The argument motivates LGAD gain as the way to raise signal-to-noise without abandoning silicon tracking.

$$\sigma_J \sim \frac{t_r}{S/N}$$



Paper 1: thickness, pixel size, and timing do not move alone



A useful contrast

- ▶ Same gain, smaller pixel: better S/N because capacitance is lower.
- ▶ In the book, the focus is on gain and d because thinner sensors help the rise time.
- ▶ Here the timing picture also runs through S/N , so a thicker detector can still look better in the total result.

My reading of it The two sources are using different timing models, so the thickness trend does not have to come out the same way.

Paper 1: the lesson I will carry forward

What I learned from the paper

More signal

gain improves S/N

- ▶ Time walk can be mitigated with methods such as ToT or CFD.
- ▶ A steep rising edge directly helps jitter.

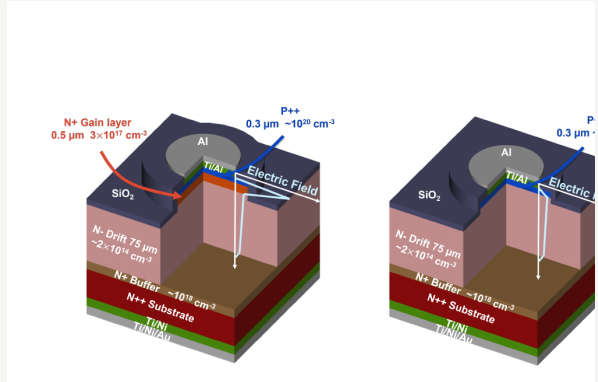
Geometry cost

pixel size and capacitance matter

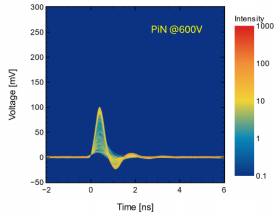
- ▶ That is why UFSD timing is a device-plus-electronics problem.

Paper 2: What I learned from this paper

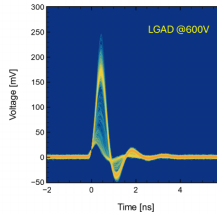
- ▶ 4H-SiC is a wide-bandgap material with high critical field and low leakage potential.
- ▶ The paper compares a fabricated LGAD stack against a PiN control stack.
- ▶ That makes the gain-layer claim testable by structure, electrical curves, and charge response.



Paper 2: what the pulse data shows



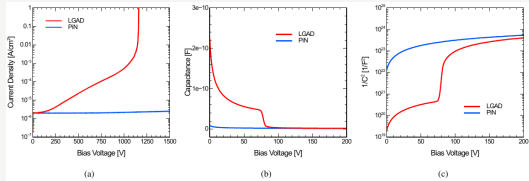
(a)



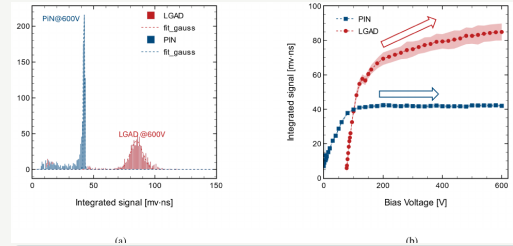
(b)

- ▶ At the same bias, the LGAD pulse is visibly larger than the PiN pulse.
- ▶ So the gain layer changes the measured signal shape, not only the DC curves.
- ▶ This is a useful bridge from structure to timing-related response.

Paper 2: what the measurements show



- ▶ Higher LGAD leakage and C-V features are consistent with the gain-layer structure.
- ▶ High reverse bias is part of the operating context in this 4H-SiC device.



Paper takeaway The measured prototype shows low-gain multiplication, but the reported gain is still only about 2–3.

Next: continue reading, then learn the tools

What I can do next

1. Read the remaining Chapter 2 sections.
2. Read more LGAD-specific papers if they are suggested.
3. Learn the Victory Device manual more carefully.
4. Try some tutorial examples in DeckBuild.
5. If possible after that, try to reproduce a known result.

Reading

finish the present theory
block



Manual

learn Victory Device basics



Tutorials

make small DeckBuild exam-
ples



Known result

attempt only when ready